

AI-Supported Research Infrastructure?

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23 April 2026

DTU rule

Du skal undgå at bruge personlige eller følsomme data i prompts uanset AI-værktøjet. Arbejder du med fortroligt materiale, skal du holde dig til de AI-værktøjer, hvor vi har en databehandlersaftale med leverandøren.

Du må altså gerne bruge AI-værktøjer, som vi ikke har en databehandlersaftale med, hvis indholdet ikke er underlagt GDPR eller på anden måde er kritisk for forretningen. Sker der brud på dette, skal du alt efter typen af informationen indmelde det til GDPR-teamet eller som en IT-sikkerhedshændelse.

Why This Matters for Research Work

Normal work

- Manuscripts, slides, grants, and reviewer responses
- Code writing, testing, and debugging
- Literature summarization and restructuring
- Overleaf, local files, Git, and co-author edits

The opportunity

- Faster writing, rewriting, and summarizing
- Better structure in papers and responses
- Faster coding support for scripts and tests
- Safer work if infrastructure is designed well

Competitive point

If productivity can move by 50–100% in parts of the research process, local infrastructure and habits matter.

From Browser Prompting to Project Agents

AI is not just a browser window.

It is also a **writing assistant**, **coding assistant**, and **project assistant**.

Browser AI

- Copy and paste workflow
- Weak project awareness
- Limited audit trail
- Easy to over-share context

Local agents

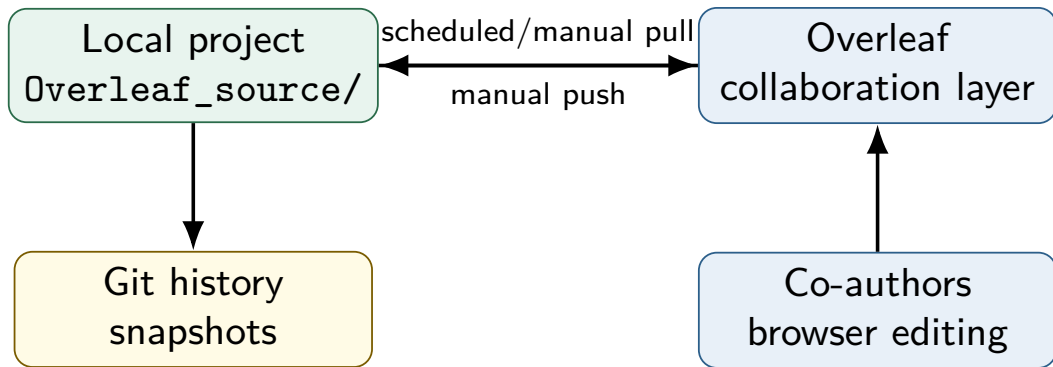
- Read and edit project files
- Run scripts, tests, and compilers
- Use logs and handovers as memory
- Work under folder and approval rules

A Generic Research Project Structure

```
Pub_XXX/  
  Overleaf_source/  
  code/  
  data/  
  Literature/  
  _ai_log.md  
  _handover.html  
  _handover.json  
  _feeders/  
  .claude/
```

- `_ai_log.md`: running diary of AI sessions.
- Handover file: onboarding brief for the next agent.
- Feeders: curated knowledge imports from related projects.
- Sensitive source data lives outside project folders in `JR/Sensitive_Data/`.

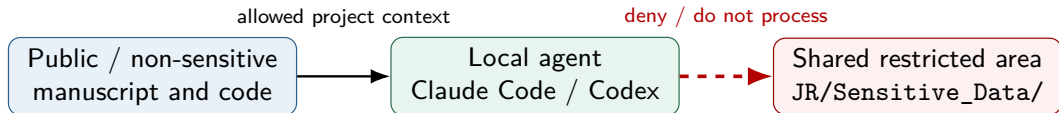
Overleaf Is Collaboration, Not the Only Copy



Design choice

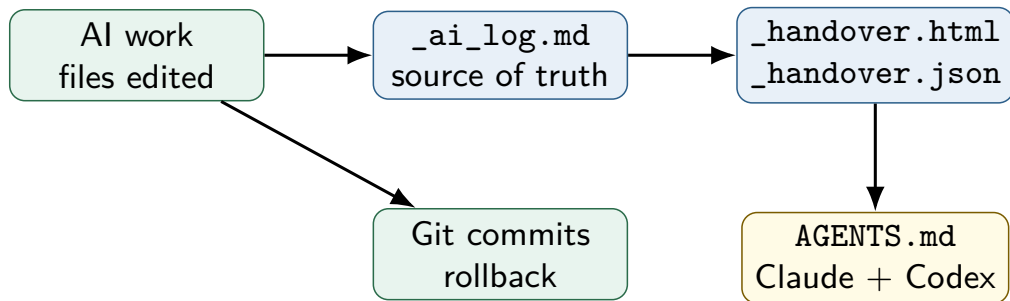
Keep Overleaf. But also keep a local, versioned copy that agents can work on and that can be backed up.

Sensitive-Data Boundary



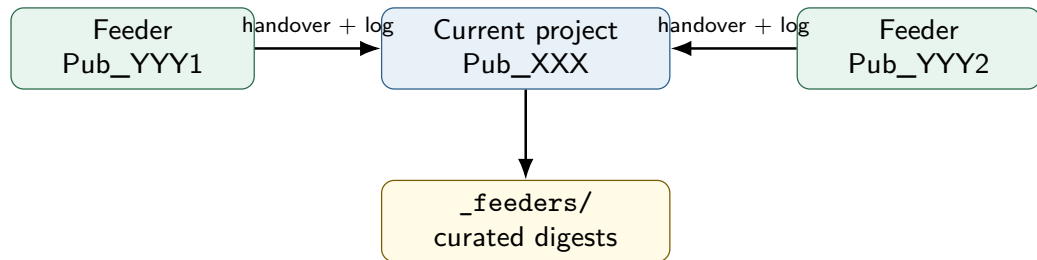
- Confidential content uses approved tools or stays outside the agent workflow.
- The safe habit is structural: keep sensitive source data outside the AI-readable project.
- Global and project deny rules create practical guardrails.
- This is safer than ad hoc copy-paste habits.

Logging, Handover, and Multi-Agent Work



- Keeps the AI context small and clean.
- Makes it easier to switch between Claude Code and Codex.
- Records what changed, which files were touched, and what comes next.

Feeder Projects: Research Memory



- Researcher decides which projects matter.
- Each feeder has its own log and handover.
- This is RAG-like, but researcher-curated and cheaper.

What Exists Now

Installation package

- Windows-first `helpi` package
- Project templates and setup scripts
- Overleaf sync and local Git
- Snapshots, rollback, and handover generation

Publication support

- Manuscript and code workflows
- Reviewer-response scaffold
- Draft loops with AI agents
- Submission package helpers

Next practical step

Test it with a few real users, clean up rough edges, and document the workflow before any broader rollout.

What DTU Could Decide

Shared infrastructure

- Project template
- Simple file classification
- Overleaf/Git synchronization
- AI log and handover format
- Sensitive-folder rules

Tools and licenses

- Serious AI coding-agent licenses
- Copilot as useful but limited
- Codex and Claude Code as project agents
- Supervisor and PhD training

Closing question

Should DTU rely on improvised browser use, or support serious AI workflows for research?